This assignment will not be graded and is only for practice.

Suppose the matrix representation of a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ with respect to ordered bases

$\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ for the domain and
$\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\}$ for the range,

is $I_{3\times3}$, i.e., the identity matrix of order 3.

Let A denote the matrix representation of the linear transformation T with respect to the standard ordered basis of $\mathbb{R}^3$ for both domain and range. Which of the following are true?

a. $A = I_{3\times3}$ i.e., identity matrix of order 3.
b. A is a singular matrix.
c. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$
d. $det(A) = 1$
e. $det(A) = -1$
f. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Solution:

Before we proceed to the solution you may want to recall the following weekly contents:

- Watch : Week 6 Tutorial 1, Week 6 Lecture 1.
- Solve : Activity 6.1 Question 8 and 9.
- Solve : Practice assignment: Question 6.

Observe that, in question, you are asked to find the matrix representation of T with respect to the standard ordered bases of $\mathbb{R}^3$ for both domain and range. So you have to find the image of each element of the standard ordered basis of $\mathbb{R}^2$ via the linear transformation T.

Note: $I_{3\times3}$ is matrix representation of the T with respect to the ordered bases β and γ.

Step 0:

1) What is $T(1, 0, 1)$? (Enter your answer in the form of a row vector in $\mathbb{R}^3$)

Note: The answer will be of the form (a, b, c) where a,b and c are in $\mathbb{R}$

No, the answer is incorrect.

Score: 0

Feedback:

From the given data, we can calculate
$T(1, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 0(1, 0, 1)$

Accepted Answers:

(Type: String) (1, 0, 0)
(Type: String) (1,0,0)

1 point

2) What is $T(0, 1, 0)$?

No, the answer is incorrect.

Score: 0

Feedback:

From the given data, we can calculate
$T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(1, 0, 1)$

Accepted Answers:

(Type: String) (0, 1, 0)
(Type: String) (0,1,0)

1 point

3) What is $T(0, 0, 1)$?

No, the answer is incorrect.

Score: 0

Feedback:

From the given data, we can calculate
$T(0, 0, 1) = 0(1, 0, 0) + 0(0, 1, 0) + 1(1, 0, 1)$

Accepted Answers:

(Type: String) (1, 0, 1)
(Type: String) (1,0,1)

1 point

Step 1:
Now we have to find the image of $(1, 0, 0)$ via the linear transformation T. We know the image of each element of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$. So at first we have to write $(1, 0, 0)$ in terms of the elements of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$.

$$(1, 0, 0) = \alpha_1(1, 0, 1) + \beta_1(0, 1, 0) + \gamma_1(0, 0, 1)$$

4) What is the value of α_1?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

5) What is the value of β_1?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

6) What is the value of γ_1?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

7) Now, can you find $T(1, 0, 0)$? **1 point**

- [ ] Yes
- [ ] No

No, the answer is incorrect.
Score: 0

Feedback:

From the given data, we can calculate
$T(1, 0, 0) = T(1(1, 0, 1) + 0(0, 1, 0) - 1(0, 0, 1))T(1, 0, 0) = 1T(1, 0, 1) + 0T(0, 1, 0) - 1T(0, 0, 1)$

Here we know the images of $(1, 0, 1)$, $(0, 1, 0)$ and $(0, 0, 1)$.
$T(1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) - 1(1, 0, 1) = (0, 0, -1)$

Accepted Answers:

Yes
No

Step 2:
Now we have to find the image of $(0, 1, 0)$ via the linear transformation T. We know the image of each element of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$. So at first we have to write $(0, 1, 0)$ in terms of the elements of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$.

$(0, 1, 0) = \alpha_2(1, 0, 1) + \beta_2(0, 1, 0) + \gamma_2(0, 0, 1)$

8) What is the value of α_2?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

9) What is the value of β_2?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

10) What is the value of γ_2?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

11) Now, can you find $T(0, 1, 0)$? **1 point**

- [ ] Yes
- [ ] No

No, the answer is incorrect.

Score: 0

Feedback:

$T(0, 1, 0) = T(0(1, 0, 1) + 1(0, 1, 0) + 0(0, 0, 1))$
$T(0, 1, 0) = 0T(1, 0, 1) + 1T(0, 1, 0) + 0T(0, 0, 1)$
$T(0, 1, 0) = (0, 1, 0)$

Accepted Answers:

Yes
No

Step 3:
Now we have to find the image of $(0, 0, 1)$ via the linear transformation T. We know the image of each element of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$. So at first we have to write $(0, 0, 1)$ in terms of the elements of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$.
$(0, 0, 1) = \alpha_3(1, 0, 1) + \beta_3(0, 1, 0) + \gamma_3(0, 0, 1)$

12) What is the value of α_3?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

13) What is the value of β_3?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

14) What is the value of γ_3?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

15) Now, can you find $T(0,0,1)$? **1 point**

- [ ] Yes
- [ ] No

No, the answer is incorrect.
Score: 0

Feedback:

$$T(0,0,1) = T(0(1,0,1) + 0(0,1,0) + 1(0,0,1))$$
$$T(0,0,1) = 0T(1,0,1) + 0T(0,1,0) + 1T(0,0,1)$$
$$T(0,0,1) = (1,0,1)$$

Accepted Answers:

Yes
No

Step 4: Now we have to find out the matrix A. For that we have to write down the images with respect to the standard ordered basis.

$$T(1,0,0) = 0\ (1,0,0) + 0\ (0,1,0) - 1\ (0,0,1)$$
$$T(0,1,0) = 0\ (1,0,0) + 1\ (0,1,0) + 0\ (0,0,1)$$
$$T(0,0,1) = 1\ (1,0,0) + 0\ (0,1,0) + 1\ (0,0,1)$$

Write the coefficients column wise to get the matrix A.

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Step 5:
In this step, you have to find the determinant of the matrix A.

16) Can you find the $det(A)$?

1 point

- [ ] Yes
- [ ] No

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes

No

Expanding with respect to the first row:

$$det(A) = 0(1-0) - 0(0-0) + 1(0+1)$$

Expanding with respect to the first column:

$$det(A) = 0(1-0) - 0(0-0) - 1(0-1)$$

You can expand with respect to any other rows or column.

17) Enter the value of **determinant of the matrix A**

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Your score is: 0/17